Erica F. Huang

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EDUCATION

Tufts University, Medford, MA **May 2027**
B.S.E. in Computer Science, Minor in Computer Engineering
3.88/4.0 GPA, Dean's List (6 semesters)

Relevant Courses: AI: Algorithms, Ethics, and Policy; Machine Learning; Machine Structure & Assembly Language Programming; Embedded Systems; Digital Circuits; Computer Organization; Technical Communication; Security; Data Structures

EXPERIENCE

MITRE*, Incoming Software Developer,* Bedford, MA **August 2027**

MITRE*, Software Engineering Intern,* Bedford, MA **Summer 2026**

- Prototype and evaluate AI/LLM integration approaches for a mission-critical Command and Control (C2) automation platform operated by the U.S. Air Force.
- Design and implement a supervisor-agent architecture that enables the chatbot to decompose complex, multi-part queries across large datasets, improving the relevance of operational planning recommendations and cutting response time in half.
- Integrate a LangChain-based AI agent with an MCP server, establishing a modular tool-calling interface that allows new capabilities to be added without modifying core agent logic.

CS Department (Tufts University)*, Teaching Assistant,* Medford, MA **May 2026 – Present**

- Tutor students and help debug in Tufts' Programming Fundamentals course.
- Plan and lead a weekly lab for students to practice what they learn in class and ask questions.

MuLIP Lab (Tufts University)*, Research Assistant,* Medford, MA **Jan 2026 – May 2026**

- Contributed to developing task, movement, and grip policies to train robots to assist humans in open-world environments.

Human-Robot Interaction Lab (Tufts University)*, Research Assistant,* Medford, MA **May 2025 – May 2026**

- Ran and measured experiments studying team collaboration to inform development of an LLM-based teamwork assistant; contributed to the resulting paper.

Cytel, *Software Engineering Intern*, Cambridge, MA **Summer 2024, Summer 2025**

- Optimized clinical trial simulation runtime by multithreading C++, bridging to R, and reading/writing through Redis
- Generated formatted simulation reports in R for client-facing trial summaries.
- Benchmarked optimization approaches – a dynamically linked library between C++ and R, parallelized C++, and parallelized R – to identify the fastest runtime.
- Presented analysis and recommendations to senior developers, informing the roadmap for further development.

FIRST Robotics Competition (FRC), *Founding Member and Coding Lead,* Wellesley, MA **Aug 2020 – May 2022**

- Led a nine-member coding team that helped build an award-winning robot.
- Expanded and diversified the coding team by gender, contributing to a second award-winning robot.
- Established and served in the team's first Director of Community Education and Outreach role.
- Created and ran summer robotics workshops for 3rd and 4th graders and partnered with local Girl Scout leaders on badge-earning events.

PROJECTS

Chatbot Agent for Command & Control Data: Inherited single-agent LangChain system during my MITRE internship and extended it to MCP server integration. The expanded data exposure revealed context-window exhaustion; redesigned as supervisor-agent architecture with specialized sub-agents to distribute workload.

Guitar Hero: Used System Verilog to program a basic version of Guitar Hero to run on an FPGA. Designed and implemented game logic, game registers, and graphics (displayed on a VGA monitor and adapter) with a group.

Universal Machine: Used C to implement a UM emulator and its memory architecture with a partner. Runs using 14 different instructions, including those that read to and write from memory, and those that load programs to run.

SKILLS

Programming Languages: C++, C, Python, Java, R, LaTeX, HTML, CSS, System Verilog, Impcore, Scheme, ARM assembly, SQL
Software: Linux, APIs, LangChain, MCP, Git, bash, zsh, VS Code, Wireshark, debugging tools, Make, Jira, Confluence, Redis
Hardware: FPGA, microcontroller, breadboard, voltmeter, oscilloscope, dev boards

ACTIVITIES

Hobbies: Snowboarding, weightlifting, basketball, triathlons, rock climbing, backpacking, logic puzzles, saxophone, LEGO
Tufts Clubs: Basketball (guard; captain), Wilderness Orientation (leader), Robotics, Mechanical Keyboards, WMFO Radio